Where irrigation exists is globally contested

2. Harmonization

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Contents

```r
# REPROJECTING AND REGRIDDING ###################################################
################################################################################

sensobol::load_packages(c("data.table", "terra", "magrittr", "here", "rnaturalearth",
                          "sf", "countrycode"))

# LOAD ALL DATASETS ############################################################

dt <- fread("./datasets/irrigated_areas/irrigated_areas.csv")

# Vector with name of datasets, used to build per-dataset rasters and stacks ---

maps <- unique(dt$dataset)

# Source all functions needed in the workflow ----------------------------------

r_functions <- list.files(path = here("functions"),
                          pattern = "\\.R$", full.names = TRUE)
lapply(r_functions, source)

# Load countries as sf (medium scale is a good balance) ------------------------

countries_sf <- ne_countries(scale = "medium", returnclass = "sf")

# Keep only what is needed -----------------------------------------------------

countries_sf <- countries_sf[, c("name", "geometry")]
```

1 Harmonization

```r
# MAKE TEMPLATES ###############################################################

# Quick check that dataset totals look plausible and match expectations --------

dt[, sum(mha), dataset] %>%
  print()
```

```
##       dataset       V1
##        <char>    <num>
##  1:   gaez_v4 287.8335
##  2:      giam 343.1581
##  3:      gmia 253.0108
##  4:      luh2 267.7412
##  5:     meier 366.2890
##  6: mirca2000 216.0765
##  7:  mirca_os 271.3082
##  8:   nagaraj 297.2691
##  9:      spam 224.7258
## 10:     gripc 313.4176
```

```r
# In terra, reprojection and aggregation are easier and safer when explicit
# template rasters defining extent, resolution and CRS are used. ---------------

template_ll_02 <- make_template_ll(0.2)
template_ll_04 <- make_template_ll(0.4)
template_ll_10 <- make_template_ll(1.0)

# Equal-area intermediate template (global Mollweide). -------------------------

# Mollweide is a global equal-area projection suitable
# for conservation-style reprojection steps.

crs_ea <- "+proj=moll +lon_0=0 +datum=WGS84 +units=m +no_defs"

# Create an equal-area template with fixed world extents and 20 km resolution ---

# We selexcted this resolution because a fixed global extent avoids shifting bounds
# 20 km is fine enough to support global reprojection without excessive computation

template_ea <- rast(xmin = -18000000, xmax = 18000000, ymin =  -9000000,
                    ymax =  9000000, crs  = crs_ea, resolution = 20000)

# BUILD 0.2º STACK #############################################################

# We selected 0.2° as a common support to preserve spatial heterogeneity and allow
# exact aggregation to coarser grids.
```

```r
# Rasterize each dataset onto the 0.2° grid (area-preserving; totals conserved).
# make_raster_ll_02() implements an area-preserving rasterization step. --------

rast_ll_list_02 <- lapply(maps, make_raster_ll_02)
mha_stack_ll_02 <- rast(rast_ll_list_02)
names(mha_stack_ll_02) <- maps

# Check totals match original (post-rasterization) -----------------------------

# To ensure that the initial conversion from table to raster does not change totals.
orig_totals <- dt[, .(total_mha_orig = sum(mha, na.rm = TRUE)), dataset]

ll02_totals <- data.table(dataset = names(mha_stack_ll_02),
                          total_mha_ll02 = global(mha_stack_ll_02, "sum",
                                                 na.rm = TRUE)[, 1])

qa_ll02 <- merge(orig_totals, ll02_totals, by = "dataset") %>%
  .[, ratio_ll02 := total_mha_ll02 / total_mha_orig] %>%
  .[order(dataset)]

# Ratio_ll02 should be 1 for all datasets (or extremely close if floating error).
qa_ll02
```

```
## Key: <dataset>
##       dataset total_mha_orig total_mha_ll02 ratio_ll02
##        <char>          <num>          <num>      <num>
##  1:   gaez_v4       287.8335       287.8335          1
##  2:      giam       343.1581       343.1581          1
##  3:      gmia       253.0108       253.0108          1
##  4:     gripc       313.4176       313.4176          1
##  5:      luh2       267.7412       267.7412          1
##  6:     meier       366.2890       366.2890          1
##  7: mirca2000       216.0765       216.0765          1
##  8:  mirca_os       271.3082       271.3082          1
##  9:   nagaraj       297.2691       297.2691          1
## 10:      spam       224.7258       224.7258          1
```

```r
# Export to .csv ---------------------------------------------------------------

fwrite(qa_ll02, "./datasets/irrigated_areas_regridded/QA_rasterization_totals_ll02.csv")

# FROM AREA TO FRACTION (0.2º LON / LAT) #######################################

# We convert to fraction because fractional coverage is bounded [0,1] and can be
# projected/resampled in a way that is conceptually consistent. It also avoids
# projecting "areas" directly in a CRS where cell areas vary with latitude.
```

```r
A_ll02_ha <- cellSize(mha_stack_ll_02, unit = "ha") # cell area in ha
f_ll02 <- (mha_stack_ll_02 * 1e6) / A_ll02_ha # Mha to ha then divide by cell area to fraction

# Clamp to [0,1] to enforce physical constraints of fractional coverage --------

CLAMP_TO_01_OLD <- TRUE
if (CLAMP_TO_01_OLD) f_ll02 <- clamp01(f_ll02)


# REPROJECT VIA EQUAL-AREA: OPERATOR TEST (bilinear vs near) ##################

# Why do we check two operators?
#   We want to quantify how much smoothing vs non-smoothing interpolation changes:
#   - totals (before renormalization)
#   - threshold classification (flip/prevalence)
#   - downstream disagreement curves

# The bilinear operator --------------------------------------------------------

rt_old_bilin <- run_old_roundtrip(f_ll02 = f_ll02,
                                  template_ea = template_ea,
                                  template_ll_02 = template_ll_02,
                                  method = "bilinear",
                                  clamp_to_01 = CLAMP_TO_01_OLD)

# The near operator ------------------------------------------------------------

rt_old_near <- run_old_roundtrip(f_ll02 = f_ll02,
                                 template_ea = template_ea,
                                 template_ll_02 = template_ll_02,
                                 method = "near",
                                 clamp_to_01 = CLAMP_TO_01_OLD)

mha_ll02_target_bilin <- rt_old_bilin$mha
mha_ll02_target_near <- rt_old_near$mha

# Quality assessment: total conservation (before renormalization) --------------
# We want to see the raw "mass loss/gain" introduced by the projection pipeline.

tot_before <- global(mha_stack_ll_02, "sum", na.rm = TRUE)[, 1]
tot_bilin <- global(mha_ll02_target_bilin, "sum", na.rm = TRUE)[, 1]
tot_near <- global(mha_ll02_target_near,  "sum", na.rm = TRUE)[, 1]

tot_check <- data.table(dataset = names(mha_stack_ll_02),
                        ratio_bilin = tot_bilin / tot_before,
                        ratio_near  = tot_near  / tot_before,
                        ratio_near_to_bilin = tot_near / tot_bilin) %>%
```

```r
  .[order(dataset)]

print(tot_check)
```

```
##       dataset ratio_bilin ratio_near ratio_near_to_bilin
##        <char>       <num>      <num>               <num>
##  1:   gaez_v4   0.9897815  0.9876177           0.9978138
##  2:      giam   1.0005444  0.9999855           0.9994414
##  3:      gmia   0.9960749  0.9936816           0.9975973
##  4:     gripc   0.9951442  0.9932962           0.9981430
##  5:      luh2   0.9727439  0.9742801           1.0015792
##  6:     meier   1.0006563  1.0011468           1.0004901
##  7: mirca2000   1.0000218  0.9981450           0.9981232
##  8:  mirca_os   0.9875943  0.9854969           0.9978762
##  9:   nagaraj   1.0024458  0.9992851           0.9968470
## 10:      spam   0.9953993  0.9938381           0.9984315
```

```r
# Renormalize both operators to match original totals exactly ------------------

# This step makes totals conservation exact and is a protection against
# the criticism that our reprojection changed totals and this is what
# explains the disagreement between cells

mha_ll02_target_bilin_cons <- renormalize_to_totals(mha_ll02_target_bilin, mha_stack_ll_02)
mha_ll02_target_near_cons  <- renormalize_to_totals(mha_ll02_target_near,  mha_stack_ll_02)

# Check exact conservation after renormalization (ratios should be 1) ----------

tot_bilin_c <- global(mha_ll02_target_bilin_cons, "sum", na.rm = TRUE)[, 1]
tot_near_c  <- global(mha_ll02_target_near_cons,  "sum", na.rm = TRUE)[, 1]

qa_total_conservation_after_renorm <- data.table(dataset = names(mha_stack_ll_02),
          ratio_bilin_cons = tot_bilin_c / tot_before,
          ratio_near_cons  = tot_near_c  / tot_before) %>%
        .[order(dataset)] %>%
  print(.)
```

```
##       dataset ratio_bilin_cons ratio_near_cons
##        <char>            <num>           <num>
##  1:   gaez_v4                1               1
##  2:      giam                1               1
##  3:      gmia                1               1
##  4:     gripc                1               1
##  5:      luh2                1               1
##  6:     meier                1               1
##  7: mirca2000                1               1
##  8:  mirca_os                1               1
##  9:   nagaraj                1               1
```

```
## 10:      spam                 1              1
```

```r
qa_total_conservation_after_renorm
```

```
##       dataset ratio_bilin_cons ratio_near_cons
##        <char>            <num>           <num>
##  1:   gaez_v4                1               1
##  2:      giam                1               1
##  3:      gmia                1               1
##  4:     gripc                1               1
##  5:      luh2                1               1
##  6:     meier                1               1
##  7: mirca2000                1               1
##  8:  mirca_os                1               1
##  9:   nagaraj                1               1
## 10:      spam                1               1
```

```r
# SELECTION OF THE MAIN HARMONIZED PRODUCT #####################################
# We choose near necause it preserves threshold-based presence/absence structure (no
# smoothing), and our quality assessment shows it minimally changes disagreement
# curves vs baseline.

mha_ll02_target <- mha_ll02_target_near_cons # MAIN
mha_ll02_target_sens_bilin <- mha_ll02_target_bilin_cons # SENSITIVITY

#Quality assessment threshold stability (before vs after) for BOTH operators ---
# We quantify whether harmonization changes binary classification at
# key tau values.

tau_set <- c(0, 100, 1000)

flip_old_bilin <- tau_flip_rates(mha_stack_ll_02,
                                 mha_ll02_target_bilin_cons,
                                 tau_ha = tau_set)

prev_old_bilin <- tau_prevalence_change(mha_stack_ll_02,
                                        mha_ll02_target_bilin_cons,
                                        tau_ha = tau_set)

flip_old_near  <- tau_flip_rates(mha_stack_ll_02,
                                 mha_ll02_target_near_cons,
                                 tau_ha = tau_set)

prev_old_near  <- tau_prevalence_change(mha_stack_ll_02,
                                        mha_ll02_target_near_cons,
                                        tau_ha = tau_set)

cat("\nFlip summary (OLD bilinear, conserved):\n")
```

```
##
## Flip summary (OLD bilinear, conserved):
```

```r
print(flip_summary(flip_old_bilin))
```

```
##    tau_ha flip_up_median flip_up_min flip_up_max flip_down_median flip_down_min
##     <num>          <num>       <num>       <num>            <num>         <num>
## 1:      0    0.051292366 0.036123489  0.05842435      0.000000000  0.000000e+00
## 2:    100    0.019483329 0.015370487  0.02729641      0.001420646  3.125735e-06
## 3:   1000    0.007424245 0.006244592  0.01164461      0.001774167  8.983361e-04
##    flip_down_max
##            <num>
## 1:  1.812926e-05
## 2:  1.609753e-03
## 3:  2.069236e-03
```

```r
cat("\nFlip summary (OLD near, conserved):\n")
```

```
##
## Flip summary (OLD near, conserved):
```

```r
print(flip_summary(flip_old_near))
```

```
##    tau_ha flip_up_median  flip_up_min flip_up_max flip_down_median
##     <num>          <num>        <num>       <num>            <num>
## 1:      0    0.001804799 0.0010489965 0.002209269     0.0018435582
## 2:    100    0.001616005 0.0010489965 0.001797923     0.0015831845
## 3:   1000    0.001024928 0.0008752057 0.001170275     0.0009830435
##    flip_down_min flip_down_max
##            <num>         <num>
## 1:   0.001065250   0.002159883
## 2:   0.001065250   0.001740409
## 3:   0.000821443   0.001147770
```

```r
cat("\nPrevalence change (OLD bilinear, conserved):\n")
```

```
##
## Prevalence change (OLD bilinear, conserved):
```

```r
print(prev_old_bilin)
```

```
##       dataset tau_ha   p_before    p_after       delta
##        <char>  <num>      <num>      <num>       <num>
##   1:  gaez_v4      0 0.07292776 0.12265945 0.049731687
##   2:  gaez_v4    100 0.04775497 0.06560292 0.017847944
##   3:  gaez_v4   1000 0.02389311 0.02959133 0.005698214
##   4:     giam      0 0.04081209 0.09901389 0.058201802
##   5:     giam    100 0.03526641 0.05491791 0.019651493
##   6:     giam   1000 0.02280036 0.02941066 0.006610303
##   7:     gmia      0 0.07823026 0.12956670 0.051336439
##   8:     gmia    100 0.04481303 0.06158760 0.016774567
```

```
##  9:      gmia   1000 0.02268971 0.02808285 0.005393142
## 10:     gripc      0 0.04998487 0.10839110 0.058406225
## 11:     gripc    100 0.03657109 0.05046686 0.013895766
## 12:     gripc   1000 0.02099493 0.02614052 0.005145584
## 13:      luh2      0 0.01479910 0.05617883 0.041379724
## 14:      luh2    100 0.01424210 0.04139660 0.027154506
## 15:      luh2   1000 0.01254232 0.02322296 0.010680635
## 16:     meier      0 0.05260299 0.10333803 0.050735048
## 17:     meier    100 0.04777560 0.06771029 0.019934685
## 18:     meier   1000 0.02904245 0.03522703 0.006184578
## 19: mirca2000      0 0.07232137 0.12421357 0.051892195
## 20: mirca2000    100 0.04337582 0.06144319 0.018067371
## 21: mirca2000   1000 0.02063735 0.02506151 0.004424165
## 22:  mirca_os      0 0.07562465 0.12687294 0.051248293
## 23:  mirca_os    100 0.04589579 0.06169200 0.015796212
## 24:  mirca_os   1000 0.02287162 0.02780904 0.004937410
## 25:   nagaraj      0 0.03697369 0.07309718 0.036123489
## 26:   nagaraj    100 0.03697369 0.05851875 0.021545063
## 27:   nagaraj   1000 0.03083225 0.03747756 0.006645312
## 28:      spam      0 0.06681570 0.11836594 0.051550239
## 29:      spam    100 0.03942989 0.05597878 0.016548889
## 30:      spam   1000 0.01902885 0.02346676 0.004437918
##        dataset tau_ha   p_before    p_after       delta
##         <char>  <num>      <num>      <num>       <num>
```

```
cat("\nPrevalence change (OLD near, conserved):\n")
```

```
##
## Prevalence change (OLD near, conserved):
```

```
print(prev_old_near)
```

```
##        dataset tau_ha   p_before    p_after         delta
##         <char>  <num>      <num>      <num>         <num>
##  1:   gaez_v4      0 0.07292776 0.07287338 -5.438778e-05
##  2:   gaez_v4    100 0.04775497 0.04781249  5.751352e-05
##  3:   gaez_v4   1000 0.02389311 0.02400189  1.087756e-04
##  4:      giam      0 0.04081209 0.04087961  6.751587e-05
##  5:      giam    100 0.03526641 0.03532080  5.438778e-05
##  6:      giam   1000 0.02280036 0.02282599  2.563102e-05
##  7:      gmia      0 0.07823026 0.07820213 -2.813161e-05
##  8:      gmia    100 0.04481303 0.04479928 -1.375323e-05
##  9:      gmia   1000 0.02268971 0.02275097  6.126440e-05
## 10:     gripc      0 0.04998487 0.05003426  4.938661e-05
## 11:     gripc    100 0.03657109 0.03658547  1.437838e-05
## 12:     gripc   1000 0.02099493 0.02114872  1.537861e-04
## 13:      luh2      0 0.01479910 0.01480473  5.626322e-06
## 14:      luh2    100 0.01424210 0.01423084 -1.125264e-05
## 15:      luh2   1000 0.01254232 0.01256170  1.937955e-05
```

```
## 16:     meier      0 0.05260299 0.05258673 -1.625382e-05
## 17:     meier    100 0.04777560 0.04777373 -1.875441e-06
## 18:     meier   1000 0.02904245 0.02902745 -1.500353e-05
## 19: mirca2000      0 0.07232137 0.07232262  1.250294e-06
## 20: mirca2000    100 0.04337582 0.04336207 -1.375323e-05
## 21: mirca2000   1000 0.02063735 0.02064485  7.501763e-06
## 22:  mirca_os      0 0.07562465 0.07558464 -4.000940e-05
## 23:  mirca_os    100 0.04589579 0.04603457  1.387826e-04
## 24:  mirca_os   1000 0.02287162 0.02306730  1.956710e-04
## 25:   nagaraj      0 0.03697369 0.03695743 -1.625382e-05
## 26:   nagaraj    100 0.03697369 0.03695743 -1.625382e-05
## 27:   nagaraj   1000 0.03083225 0.03079286 -3.938426e-05
## 28:      spam      0 0.06681570 0.06677382 -4.188484e-05
## 29:      spam    100 0.03942989 0.03947053  4.063455e-05
## 30:      spam   1000 0.01902885 0.01908261  5.376263e-05
##       dataset tau_ha   p_before    p_after         delta
##        <char>  <num>      <num>      <num>         <num>
```

```
# SAVE QUALITY ASSESSMENT OUTPUTS FOR SUPPLEMENTARY MATERIALS #################

# These tables are evidence that harmonization does not bias results -----------

fwrite(
  qa_total_conservation_after_renorm,
  "./datasets/irrigated_areas_regridded/QA_total_conservation_after_renormalization.csv"
)

fwrite(flip_old_bilin, "./datasets/irrigated_areas_regridded/QA_flip_rates_old_bilinear.csv")
fwrite(prev_old_bilin, "./datasets/irrigated_areas_regridded/QA_prevalence_change_old_bilinear
fwrite(flip_old_near,  "./datasets/irrigated_areas_regridded/QA_flip_rates_old_near.csv")
fwrite(prev_old_near,  "./datasets/irrigated_areas_regridded/QA_prevalence_change_old_near.csv

# Sanity check to guarantee that the output is actually on the intended grid
# for downstream analysis ------------------------------------------------------

res(mha_ll02_target)
```

```
## [1] 0.2 0.2
```

```
# Verify totals after full round-trip vs ORIGINAL dt totals --------------------
# Another reviewer-proof check: final harmonized rasters still match
# original totals.

ll02_target_totals <- data.table(dataset = names(mha_ll02_target),
                                 total_mha_ll02_target = global(mha_ll02_target,
                                                                "sum", na.rm = TRUE)[, 1])

check_02 <- merge(orig_totals, ll02_target_totals, by = "dataset") %>%
  .[, ratio := total_mha_ll02_target / total_mha_orig] %>%
```

```
  .[order(dataset)]

print(check_02)
```

```
## Key: <dataset>
##       dataset total_mha_orig total_mha_ll02_target ratio
##        <char>          <num>                 <num> <num>
##  1:   gaez_v4       287.8335              287.8335     1
##  2:      giam       343.1581              343.1581     1
##  3:      gmia       253.0108              253.0108     1
##  4:     gripc       313.4176              313.4176     1
##  5:      luh2       267.7412              267.7412     1
##  6:     meier       366.2890              366.2890     1
##  7: mirca2000       216.0765              216.0765     1
##  8:  mirca_os       271.3082              271.3082     1
##  9:   nagaraj       297.2691              297.2691     1
## 10:      spam       224.7258              224.7258     1
```

```
# SAVE THE ALIGNED 0.2º STACK #################################################

# We savethe harmonized multi-layer raster for reproducibility and reuse.

writeRaster(mha_ll02_target, filename = "mha_stack_ll_02_aligned_NEAR.tif",
            overwrite = TRUE, gdal = c("COMPRESS=LZW"))

# SAVE THE OTHER RASTERS AS WELL ################################################

writeRaster(mha_stack_ll_02, "mha_stack_ll_02_baseline.tif", overwrite=TRUE, gdal="COMPRESS=LZW

writeRaster(mha_ll02_target_bilin_cons, "mha_stack_ll_02_aligned_BILIN.tif", overwrite=TRUE, gd

# Export 0.2° long table -------------------------------------------------------

irrigated_areas_regridded_02 <- as.data.table(
  as.data.frame(mha_ll02_target, xy = TRUE, na.rm = FALSE))

irrigated_areas_regridded_02 <- melt(irrigated_areas_regridded_02,
                                     id.vars = c("x", "y"),
                                     variable.name = "dataset",
                                     value.name = "mha",
                                     variable.factor = FALSE)

setnames(irrigated_areas_regridded_02, c("x", "y"), c("lon", "lat"))


# AGGREGATE TO 0.4° ############################################################

# Spatial aggregation is a conservative robustness test: it reduces fine-scale
```

```r
# misalignment/noise so disagreement persisting at coarser scales is
# harder to dismiss. fact=2 maps 0.2° to 0.4°; summing preserves total irrigated area.

mha_stack_ll_04 <- aggregate(mha_ll02_target, fact = 2, fun = sum, na.rm = TRUE)
res(mha_stack_ll_04)
```

```
## [1] 0.4 0.4
```

```r
# Check totals preserved under aggregation ------------------------------------

orig_totals_stack <- global(mha_ll02_target, "sum", na.rm = TRUE)[, 1]
agg04_totals <- global(mha_stack_ll_04, "sum", na.rm = TRUE)[, 1]
qa_04 <- data.table(dataset = names(mha_ll02_target),
                    ratio = agg04_totals / orig_totals_stack)

# Export assessment -----------------------------------------------------------

fwrite(qa_04,
       "./datasets/irrigated_areas_regridded/QA_aggregation_totals_04deg.csv")

# Export 0.4° long table ------------------------------------------------------

irrigated_areas_regridded_04 <- as.data.table(
  as.data.frame(mha_stack_ll_04, xy = TRUE, na.rm = FALSE)) %>%
  melt(id.vars = c("x", "y"), variable.name = "dataset", value.name = "mha",
       variable.factor = FALSE)

setnames(irrigated_areas_regridded_04, c("x", "y"), c("lon", "lat"))

# AGGREGATE TO 1° #############################################################

# Even stronger conservative scale; if disagreement persists at 1°, it is not a
# fine-resolution alignment artefact. fact=5 maps 0.2° to 1°.

mha_stack_ll_10 <- aggregate(mha_ll02_target, fact = 5, fun = sum, na.rm = TRUE)
res(mha_stack_ll_10)
```

```
## [1] 1 1
```

```r
# Check totals preserved under aggregation ------------------------------------

agg10_totals <- global(mha_stack_ll_10, "sum", na.rm = TRUE)[, 1]
qa_10 <- data.table(dataset = names(mha_ll02_target),
                    ratio = agg10_totals / orig_totals_stack)

# Export quality assessment ---------------------------------------------------

fwrite(qa_10, "./datasets/irrigated_areas_regridded/QA_aggregation_totals_10deg.csv")
```

```r
# Export 1º long table ----------------------------------------------------------

irrigated_areas_regridded_10 <- as.data.table(
  as.data.frame(mha_stack_ll_10, xy = TRUE, na.rm = FALSE)) %>%
  melt(id.vars = c("x", "y"), variable.name = "dataset", value.name = "mha",
       variable.factor = FALSE)

setnames(irrigated_areas_regridded_10, c("x", "y"), c("lon", "lat"))

# ADD COUNTRIES + CONTINENT AND EXPORT #########################################
# This step enables regional summaries and continent/country decomposition of
# disagreement. drop_all_zero_or_na_cells() reduces file size and speeds
# later computations by removing cells that contain no irrigation in all datasets.

# 0.2º --------------------------------------------------------------------------

irrigated_areas_regridded_02 <- add_admin(irrigated_areas_regridded_02)
```

```
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
```

```r
irrigated_areas_regridded_02 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_02)

# Final total check after dropping cells ---------------------------------------
# Even after dropping "all-zero" rows, dataset totals should remain unchanged.

merge(irrigated_areas_regridded_02[, .(current = sum(mha, na.rm = TRUE)), dataset],
      orig_totals, by = "dataset") %>%
  .[, fraction := current / total_mha_orig] %>%
  .[order(dataset)] %>%
  print()
```

```
## Key: <dataset>
##       dataset  current total_mha_orig  fraction
##        <char>    <num>          <num>     <num>
##  1:   gaez_v4 285.5135       287.8335 0.9919400
##  2:      giam 339.7342       343.1581 0.9900224
##  3:      gmia 249.8387       253.0108 0.9874624
##  4:     gripc 309.9096       313.4176 0.9888073
##  5:      luh2 265.5824       267.7412 0.9919371
##  6:     meier 362.3263       366.2890 0.9891815
##  7: mirca2000 213.6309       216.0765 0.9886819
##  8:  mirca_os 268.6011       271.3082 0.9900221
##  9:   nagaraj 293.5383       297.2691 0.9874498
## 10:      spam 222.3060       224.7258 0.9892322
```

```r
# Export 0.2º file --------------------------------------------------------------
```

```r
fwrite(irrigated_areas_regridded_02,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_02.csv")

# 0.4° ------------------------------------------------------------------------

irrigated_areas_regridded_04 <- add_admin(irrigated_areas_regridded_04)
```

```
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
```

```r
irrigated_areas_regridded_04 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_04)

fwrite(irrigated_areas_regridded_04,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_04.csv")

# 1° --------------------------------------------------------------------------

irrigated_areas_regridded_10 <- add_admin(irrigated_areas_regridded_10)
```

```
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
```

```r
irrigated_areas_regridded_10 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_10)

fwrite(irrigated_areas_regridded_10,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_10.csv")
```

1.1 Does harmonization change the disagreement curves?

```
# DOES HARMONIZATION CHANGE THE DISAGREEMENT CURVES? ##########################
#
# The main results of the paper are disagreement curves: the fraction of grid
# cells that disagree on irrigation presence as a function of the detectability
# threshold tau, together with their decomposition into existential vs marginal,
# minor vs major disagreement, etc.
#
# Even if we have shown that:
#   - global irrigated-area totals are conserved, and
#   - irrigation prevalence is stable under nearest-neighbour reprojection,
# this does not automatically guarantee that the shape of the disagreement
# curves is unaffected by harmonization.
#
# In this QA we therefore explicitly test whether the harmonization step
# (reprojection + resampling) biases disagreement metrics across tau by:
#   1) computing disagreement curves on:
#        (a) the baseline aligned 0.2° stack (mha_stack_ll_02),
#       (b) the nearest-neighbour round-trip, renormalized (mha_ll02_target_near_cons),
#      (c) the bilinear round-trip, renormalized (mha_ll02_target_bilin_cons);
#   2) comparing curves (b) and (c) against the baseline across the full tau grid;
#   3) summarizing for each key metric the maximum absolute change across tau.
#
# If near_minus_base deltas are c. 0 for all tau, harmonization does not distort
# disagreement results. If bilin_minus_base deltas are large at low tau, this
# indicates that bilinear smoothing inflates the spatial footprint of small
# irrigation signals and artificially increases disagreement near the detection
# limit.

# DEFINE THE SAME TAU GRID USED IN THE ANALYSIS ###############################

# Define a dense grid of tau values (in ha) from 1 ha (10^0) to 1e5 ha (10^5),
# on a log10 scale with step 0.1. tau = 0 (label "any_positive") is added to
# represent the "any non-zero irrigation" case used in the main analysis.

log_tau_min <- 0
log_tau_max <- 5
log_step <- 0.1

taus_ha_nonzero <- 10^(seq(log_tau_min, log_tau_max, by = log_step))
taus_ha <- c(0, taus_ha_nonzero)
taus_mha <- taus_ha * 1e-6

# Labels: "any_positive" for tau = 0, then e.g. "1ha", "1.26ha", ..., "100000ha"
tau_labels <- ifelse(taus_ha == 0, "any_positive",
                     sprintf("%gha", signif(taus_ha, 3)))
```

```r
tau_grid <- data.table(tau_label = tau_labels,
                       tau_ha = taus_ha,
                       tau_mha = taus_mha)

# RELOAD PRODUCTS FOR THE QUALITY CHECK ########################################

# Reload from disk to avoid invalid external pointers during knitting.

mha_stack_ll_02 <- rast("mha_stack_ll_02_baseline.tif")
mha_ll02_target_near_cons <- rast("mha_stack_ll_02_aligned_NEAR.tif")
mha_ll02_target_bilin_cons <- rast("mha_stack_ll_02_aligned_BILIN.tif")

# Build disagreement curves for each product -----------------------------------
# compute_curve_from_stack() applies the tau-based classification and disagreement
# metrics used in the main analysis, returning e.g. frac_disagree(tau), etc.

curve_base <- compute_curve_from_stack(mha_stack_ll_02, tau_grid) %>%
  .[, product:= "base_ll02"]

curve_near <- compute_curve_from_stack(mha_ll02_target_near_cons, tau_grid) %>%
  .[, product:= "rt_near_cons"]

curve_bilin <- compute_curve_from_stack(mha_ll02_target_bilin_cons, tau_grid) %>%
  .[, product:= "rt_bilin_cons"]

curves <- rbindlist(list(curve_base, curve_near, curve_bilin), use.names = TRUE)
setorder(curves, product, tau_mha)

# Compute deltas vs baseline for key metrics -----------------------------------

# For each tau and metric, we compute:
#   $\Delta$_near(tau)  = metric_near(tau)  - metric_base(tau)
#   $\Delta$_bilin(tau) = metric_bilin(tau) - metric_base(tau)
# A robust harmonization should give $\Delta$_near(tau) = 0 for all tau.

metrics <- c("frac_disagree", "frac_existence", "frac_marginal",
             "frac_minor_disagree", "frac_major_disagree",
             "share_existence", "share_marginal")

base_keep <- curve_base[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE]

delta_near <- merge(
  curve_near[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE],
  base_keep,
  by = c("tau_label", "tau_ha", "tau_mha"),
  suffixes = c("_near", "_base"))
```

```r
delta_bilin <- merge(
  curve_bilin[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE],
  base_keep,
  by = c("tau_label", "tau_ha", "tau_mha"),
  suffixes = c("_bilin", "_base"))

# Convert to delta columns (harmonized - baseline) -----------------------------

for (m in metrics) {
  delta_near[, (m):= get(paste0(m, "_near")) - get(paste0(m, "_base")) ]
  delta_bilin[, (m):= get(paste0(m, "_bilin")) - get(paste0(m, "_base")) ]
}

delta_near[, comparison:= "near_minus_base"]
delta_bilin[, comparison:= "bilin_minus_base"]

delta_all <- rbindlist(list(delta_near, delta_bilin), use.names = TRUE, fill = TRUE)

# Summarize worst-case changes across tau for headline metrics -------------------
# For the key metrics (frac_disagree, frac_existence, frac_marginal), we
# extract, for each comparison, the tau at which |$\Delta$(tau)| is largest and the
# corresponding $\Delta$(tau). This yields a compact "worst-case impact" summary.

summary_near <- rbindlist(lapply(
  c("frac_disagree", "frac_existence", "frac_marginal"),
  function(m) summarize_delta(delta_near, m)
))[, comparison:= "near_minus_base"]

summary_bilin <- rbindlist(lapply(
  c("frac_disagree", "frac_existence", "frac_marginal"),
  function(m) summarize_delta(delta_bilin, m)
))[, comparison:= "bilin_minus_base"]

delta_summary <- rbindlist(list(summary_near, summary_bilin), use.names = TRUE)
setorder(delta_summary, comparison, metric)

# Interpretation:
#   - If near_minus_base deltas are ~0 for all tau (and max |$\Delta$| is O(10^-3) or less),
#     the disagreement analysis is robust to nearest-neighbour harmonization.
#   - If bilin_minus_base deltas are large at low tau (especially tau = 0), bilinear
#     interpolation artificially inflates the footprint of small irrigation values
#     and increases disagreement near the detection limit.

# Nearest-neighbour harmonization does not alter the disagreement curves in
# any meaningful way and that is why we have used nearest-neighbour as the
# interpolation in the main analysis - disagreement is not an harmonization effect.
print(delta_summary)
```

```
##            metric max_abs_delta tau_ha_at_max tau_label_at_max  delta_at_max
##            <char>         <num>         <num>           <char>         <num>
## 1:  frac_disagree  0.0332265432        0.0000     any_positive  0.0332265432
## 2: frac_existence  0.0332265432        0.0000     any_positive  0.0332265432
## 3:  frac_marginal  0.0192117284      794.3282            794ha  0.0192117284
## 4:  frac_disagree  0.0004796296    50118.7234          50100ha -0.0004796296
## 5: frac_existence  0.0001376543      630.9573            631ha  0.0001376543
## 6:  frac_marginal  0.0004524691    50118.7234          50100ha -0.0004524691
##          comparison
##              <char>
## 1: bilin_minus_base
## 2: bilin_minus_base
## 3: bilin_minus_base
## 4:  near_minus_base
## 5:  near_minus_base
## 6:  near_minus_base
```

```r
# Save tables for supplementary materials --------------------------------------

fwrite(curves,
       "./datasets/irrigated_areas_regridded/QA_disagreement_curves_base_near_bilin.csv")
fwrite(delta_all,
       "./datasets/irrigated_areas_regridded/QA_disagreement_curve_deltas_vs_base.csv")
fwrite(delta_summary,
       "./datasets/irrigated_areas_regridded/QA_disagreement_delta_summary.csv")
```

2 Session information

```r
# SESSION INFORMATION ##########################################################

sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framew
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
```

```
##
## other attached packages:
## [1] countrycode_1.6.1   sf_1.1-0            rnaturalearth_1.1.0
## [4] here_1.0.2          magrittr_2.0.4      terra_1.8-93
## [7] data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] cli_3.6.5          knitr_1.51         rlang_1.1.7        xfun_0.55
##  [5] otel_0.2.0         DBI_1.3.0          KernSmooth_2.23-26 rprojroot_2.1.1
##  [9] htmltools_0.5.9    e1071_1.7-17       rmarkdown_2.30     grid_4.5.2
## [13] evaluate_1.0.5     classInt_0.4-11    fastmap_1.2.0      yaml_2.3.12
## [17] compiler_4.5.2     codetools_0.2-20   Rcpp_1.1.1         rstudioapi_0.17.1
## [21] digest_0.6.39      class_7.3-23       tools_4.5.2        proxy_0.4-29
## [25] units_1.0-0
```

```r
## Return the machine CPU ---------------------------------------------------

cat("Machine:     "); print(benchmarkme::get_cpu()$model_name)
```

```
## Machine:
## [1] "Apple M1 Max"
```

```r
## Return number of true cores ------------------------------------------------

cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
## [1] 10
```

```r
## Return number of threads ---------------------------------------------------

cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
## [1] 10
```